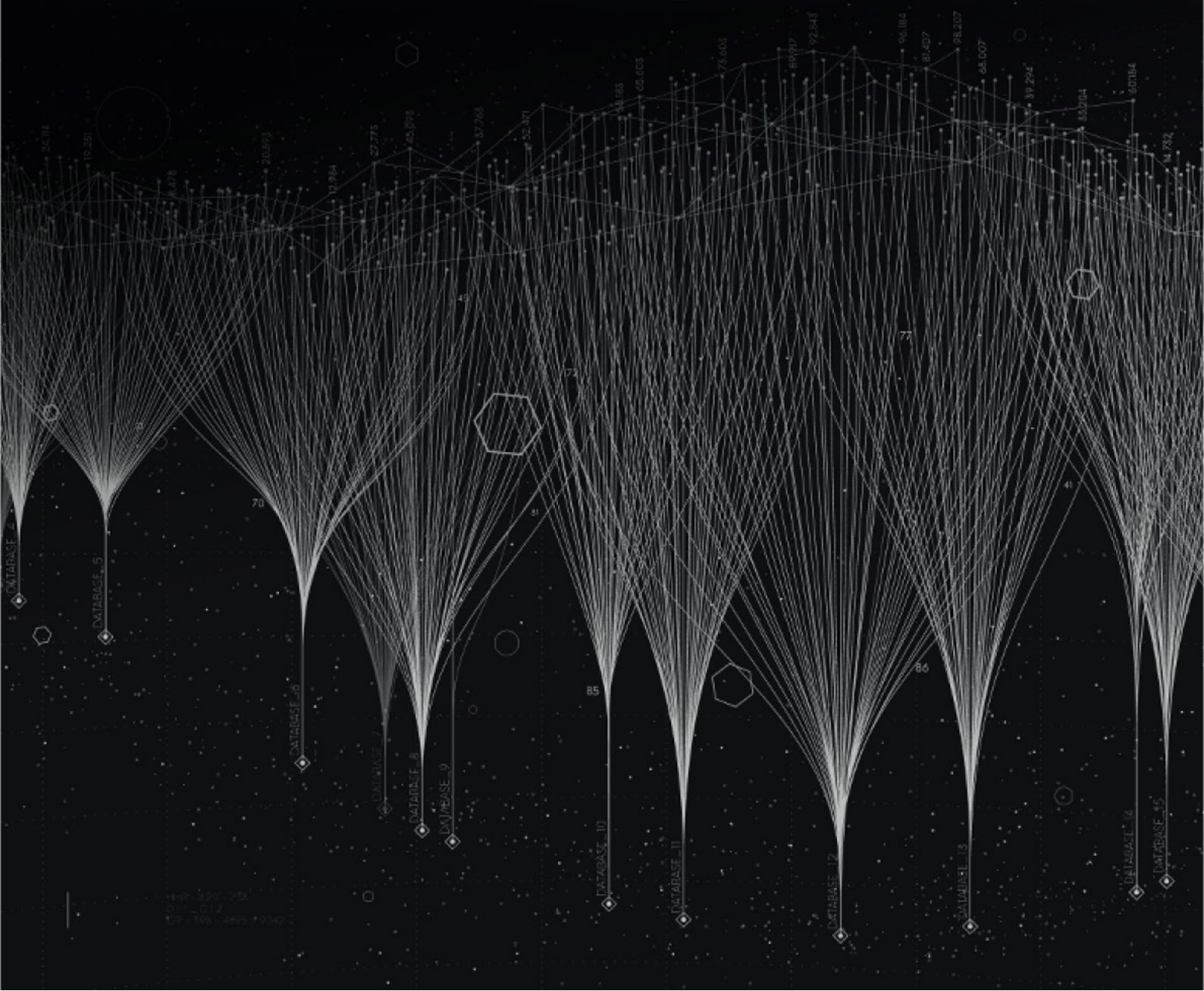
Professional Education

Forecasting Technology Innovation:

Using Data for Strategic Advantage





There's a clear necessity to be able to measure and forecast rates of change in technology in order to anticipate which innovations are poised to become consequential in the marketplace.

Prof. Jessika Trancik

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Why Is Forecasting Technology Innovation Important



Decision makers who are skilled at forecasting technology innovation can have a competitive advantage.

Investors can stay ahead of industry trends and build a successful technology investment portfolio.

Scientists and engineers can better prioritize resources invested in technology research and development.

Companies can be more successful at anticipating and entering new and promising markets.

Governments can support national competitiveness and advance technology for the benefit of society.

The better these decision makers are at measuring and forecasting technology innovation, the more effective their decision-making processes around technology investment can be.

Forecasting technology innovation for a competitive advantage

"Often forecasts stop at predicting what will happen. Go one step further to consider the implied impact on how companies will compete. Think about how your company could secure advantage and ways to adapt your strategy to match."

(Reeves, M., Ramaswamy, S., & O'Dea, A., 2022)

"Measuring innovation is key to achieving tangible results. Establishing viable innovation metrics is essential to bringing leaders on board, lining up the resources and attracting the talented colleagues you need to turn ideas into done deals.''

(Tucker, R., Milliken, K., & Kirsner, S., 2021)

"The larger the number of people who can make data-driven global decisions, the better futures we can implement together."

(Börner, K., Rouse, W. B., Trunfio, P., & Stanley, H. E., 2018)

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About this Course



MIT Professional Education's online course **Forecasting Technology Innovation: Using Data for Strategic Advantage** explores the data informed process of measuring, predicting, and influencing technological innovation.

1. Measuring technological innovation
2. Predicting technological innovation
3. Influencing technological innovation

As you gain mastery in these skills, you will discover how to apply them in your sector to make better decisions about technology investment and design.

8 weeks | 8 - 10 hours a week | Online

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Key Takeaways

The goal of this course is to equip professionals and organizations in both the public and private sectors with the data-based knowledge and tools to drive better technology investment and design decisions. Throughout this course, you will:

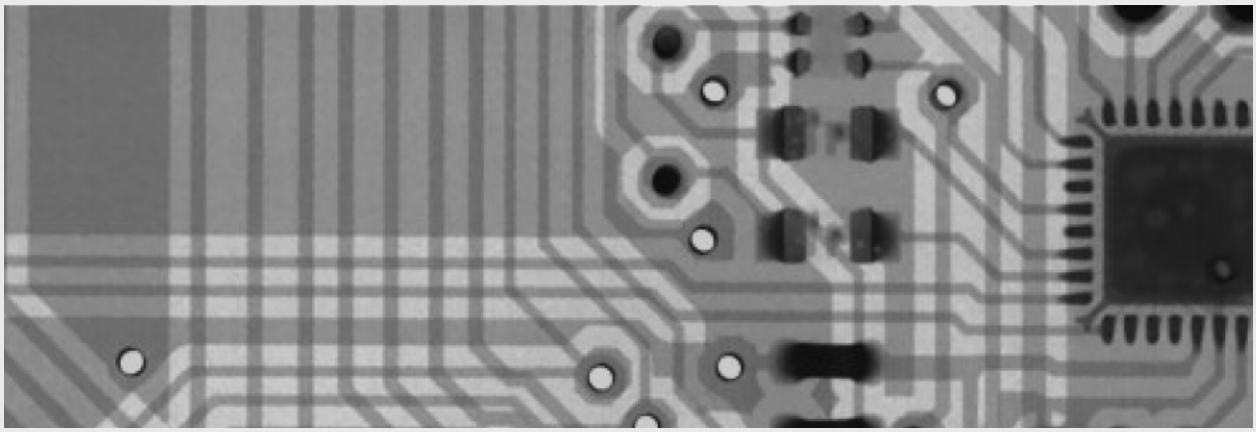
01 **Develop an understanding** of how large data sets can be used to gain insights into the dynamics of technological innovation.

02 **Learn how to compare the rates of progress** of various technologies and products.

03 **Explore the state of the art in theories** of technological innovation and their application.

04 **Discover how to apply data analysis** and theory to guide technology investment and design decisions.

05 **Gain insights into how to improve decision-making** to support technological innovation when designing financial portfolios, research and development portfolios, and public policy.



Industry Applications

Any organization looking to gain a competitive advantage by improving existing technologies and implementing new ones can benefit from an advanced, data-informed decision-making process. This course will be particularly relevant to:



Technology innovation is essential to the evolving healthcare industry. New technologies involving AI, data science, and analytics are poised to improve various facets of healthcare. For example, to eliminate future supply chain disruptions 50% of life science companies and healthcare providers are set to boost investments in AI and advanced analytics (Limaye, 2021). However, questions remain about how to successfully implement these new technologies within existing processes, and which technologies to prioritize. Successful technology innovation strategies promise to further understanding of the genetic factors behind disease, optimize clinical studies and trials, and streamline regulatory approval processes.

86% of manufacturers in the US have stated that smart factories and their associated technologies would be the most important drivers of competition by 2025 (Laaper et al., 2020). The extensive list of potential supporting technologies for this transition— including IoT, data analytics, cloud computing, and machine learning and AI, among others—highlights the need for a data-driven, selective approach to technology investment and design in a sector marked by heightened competition. If equipped with the expertise and tools necessary to predict and influence the evolution of these technologies, organizations stand to gain a substantial advantage over their competitors.





Investing in technology innovation means working with uncertainty. While no crystal ball exists, data-driven forecasting models and mechanistic models of innovation can point to fast improving industries and particularly promising technologies within them. This information is essential for balancing portfolios of investments to achieve success. Whether investments are being made in early-stage technologies or more advanced ones, it is important to recognize both trends and noise in the development of technologies to appropriately diversify or concentrate investments. Using data-informed methods to measure and forecast technological change, investment firms can gain a significant advantage

The global energy market is experiencing a rapid transition driven by fundamental changes in technological development, regulations, consumer preferences, and investor sentiments. The evergrowing pressure to reduce greenhouse gas emissions through domestic and international policy and through corporate sustainability programs is having a significant impact on the choices of technologies that serve the energy sector. Simultaneously, the shifting of energy portfolios through increased electrification, emerging alternate fuel and energy storage options, and efficiency gains are further advancing the need for new and innovative technology solutions. The challenge is to be armed with the best tools to understand and act on the most effective energy options to meet short and long-term market demands. (McKinsey, 2022).





The chemical industry is directly or indirectly involved in the manufacturing of nearly every product, and is currently undergoing a shift as its business processes and models become digitized. Concurrently, calls for increased circular economy practices and more sustainable operations are emerging from an array of industries, offering the chemical industry major new opportunities for growth. On the other hand, each individual organization will need to be prepared to take a proactive—rather than reactive—approach to technology innovation and implementation, using available data to weigh opportunities and risks; make effective, informed decisions; and gain a competitive advantage. Data-informed models can help businesses benefit from advances in technology, including through improved supply chain management, preventative maintenance, and other cost-reduction strategies.

With the ability to measure and track technological progress, public policymakers can analyze and promote technology innovations in terms of societal goals. This ability to understand and influence the drivers behind technology advancements helps to remove the guesswork from the process of choosing and developing the right technologies, offering governments insights into how money and time can be invested to best meet their objectives.



Participant Profile

Although the expertise found throughout this course can be applied to a wide series of industries, it is especially beneficial for those working in sectors such as chemicals, life sciences, manufacturing, investment, energy, and public policy. Though not limited to any job level, the most common roles that will benefit from the course include:

Research and development managers looking to advance their organization's technology portfolio decision-making through data-informed insights.

Production or manufacturing operations managers who wish to incorporate data collection and analysis to anticipate technological changes and prioritize their investments in innovation

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C-suite and executive level management in technology-related firms in charge of establishing a data-informed technology plan for their organization.

Public policy makers from technology-related sectors who want to discover how technological progress can be measured and predicted to help meet societal goals.

Private investors interested in optimizing technology-related portfolios based on technology performance data, allowing them to improve technology investment decisions.

Instructor



Prof. Jessika Trancik

Professor in the Institute for Data, Systems, and Society, MIT

Jessika Trancik is a professor in the Institute for Data, Systems, and Society at the Massachusetts Institute of Technology. Her research examines the impacts of technologies and the reasons behind technological change.

She has developed theory and predictive models to understand why some technologies improve faster than others, and what technology features enable rapid innovation. Trancik has developed models for forecasting technological change, which inform engineering design, public policy, and investment portfolios. Several of her theories and models have been applied to new and developing energy technologies, such as solar energy and batteries, and to electricity and transportation systems. Her models have also been used to inform government innovation policy, and applied in diverse industries, including finance, healthcare, manufacturing, software, and consumer products. Her work has been published in journals such as Nature, Proceedings of the National Academy of Sciences, Nature Energy, Nature Climate Change, and Environmental Science and Technology, and has been featured by news outlets such as the New York Times, Washington Post, Financial Times, and NPR.

> "There's a clear necessity to be able to measure and forecast rates of change in technology in order to anticipate which innovations are poised to become consequential in the marketplace."

Professor Trancik received her B.S. from Cornell University and her Ph.D. from the University of Oxford as a Rhodes Scholar.

She has developed theory and predictive models to understand why some technologies improve faster than others, and what technology features enable rapid innovation.

Her models have also been used to inform government innovation policy, and applied in diverse industries, including finance, healthcare, manufacturing, software, and consumer products

Course Outline

Module 1
Course overview and motivation

1. Motivation
2. Reflections on forecasting
3. Overview and background

Module 2
Definitions and performance metrics

1. Defining technology
2. Technological sophistication and progress Performance metrics: Measuring cost, time, or other resources per service provided
3. Consequences: The beneficial and harmful
4. impacts of technology

- Assignment 1: Performance metrics
- Assignment 2: Price indices
- Assignment 3: Data analysis
- Assignment 4: Measuring technology's benefits and harms

Module 3
Data, trends, and data-driven models

1. Examples of change in technology performance over time
2. Data-driven models of technology innovation

- Assignment 1: Dataset analysis

Module 4
Forecasting using data-driven models

1. Forecasting methods: From expert elicitation to data-driven models
2. Data-driven forecasting: Accounting for uncertainty
3. Discussion of technology forecasting

- Assignment 1: Interpreting an application of Moore's Law
- Assignment 2: Analyzing a data source as a basis for forecasting

Module 5
Mechanistic models part 1: Innovation of interconnected components

1. Building intuition
2. Mechanistic model example: Innovation of interconnected components

- Assignment 1: Visualizing component dependency

Module 6.
Mechanistic models part 2: Innovation mechanisms

1. The multi-level drivers model
2. Examples

- Assignment 1: Assessing technology innovation policy

Module 7
Portfolio optimization

1. What is a technology portfolio problem?
2. How innovation forecasts can inform technology portfolio decisions
3. Solving technology portfolio problems

- Assignment 1: Vaccine development (Optional)
- Assignment 2: Portfolio problem

Module 8
Applications of data-informed insights to technology decisions

1. Brief review of course material
2. Application scenarios
3. Concluding remarks

- Assignment 1: Application scenarios

Certificate of Completion

All participants who successfully complete the course will receive an official certificate from MIT Professional Education. In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

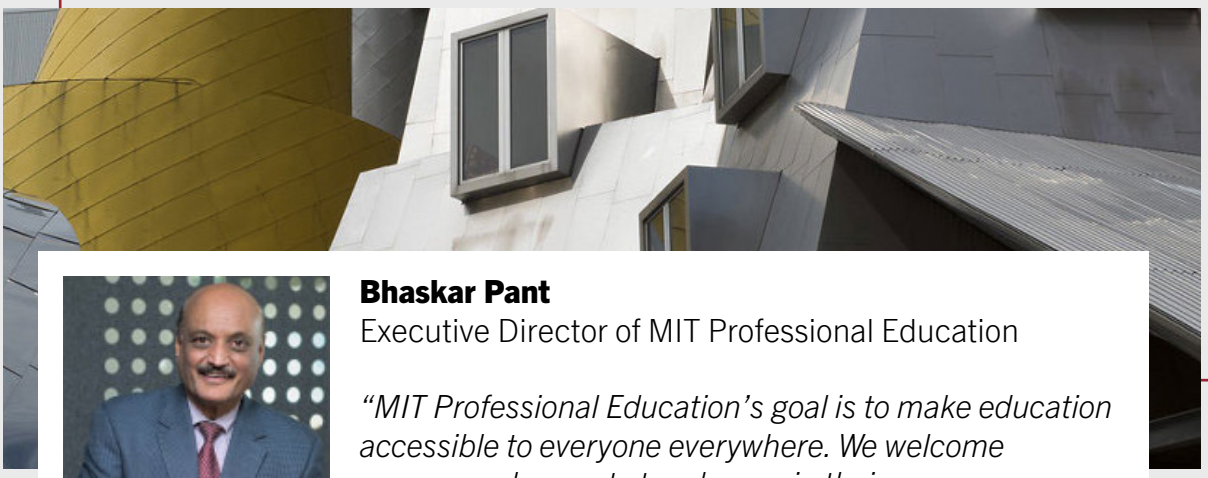
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

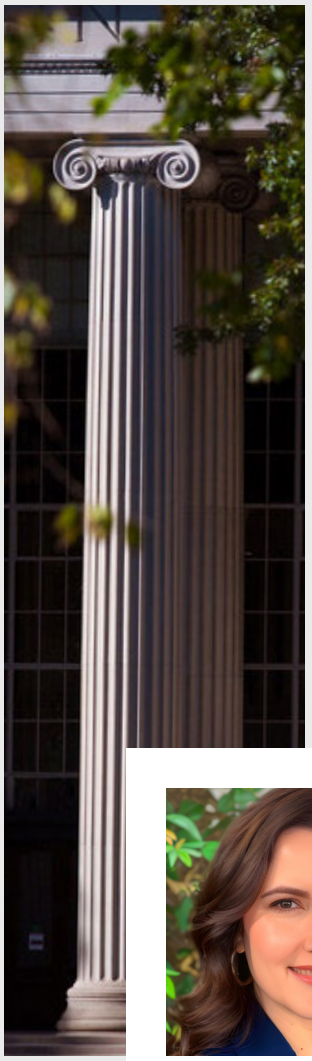


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is revolutionizing the digital learning experience. In fact, we are no longer talking about online-only, but rather an interactive and collaborative learning experience that's digitally powered: a Beyond Online experience.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Senior Director of Global Programs, Digital Plus Programs, and Corporate Programs

"Forecasting technology innovation is a crucial approach for tackling technological advancements within rapidly evolving market landscapes and is important to enhance competitiveness of organizations."

MIT Professional Education

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Benefits of Joining the Professional Alumni Community



MIT Professional Education offers a number of benefits for participants who successfully complete our courses:

1. Digital Certificate of Completion and MIT CEUs*.
2. 15% benefit for online programs and short-duration, in-person programs offered by MIT Professional Education.
3. Access to the virtual campus for 6-months after program completion.
4. Exclusive access to the MIT Professional Education Alumni LinkedIn group, connecting you with a global network of fellow MIT Professional Education alumni.

*To obtain CEUs, complete the accreditation confirmation, which is available at the end of the course. CEUs are calculated for each course based on the number of learning hours.

MIT Professional Education

Are you prepared to create innovative solutions?

Frequently Asked Questions

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Website: **https://professionalprograms.mit.edu**

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